



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO S M FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

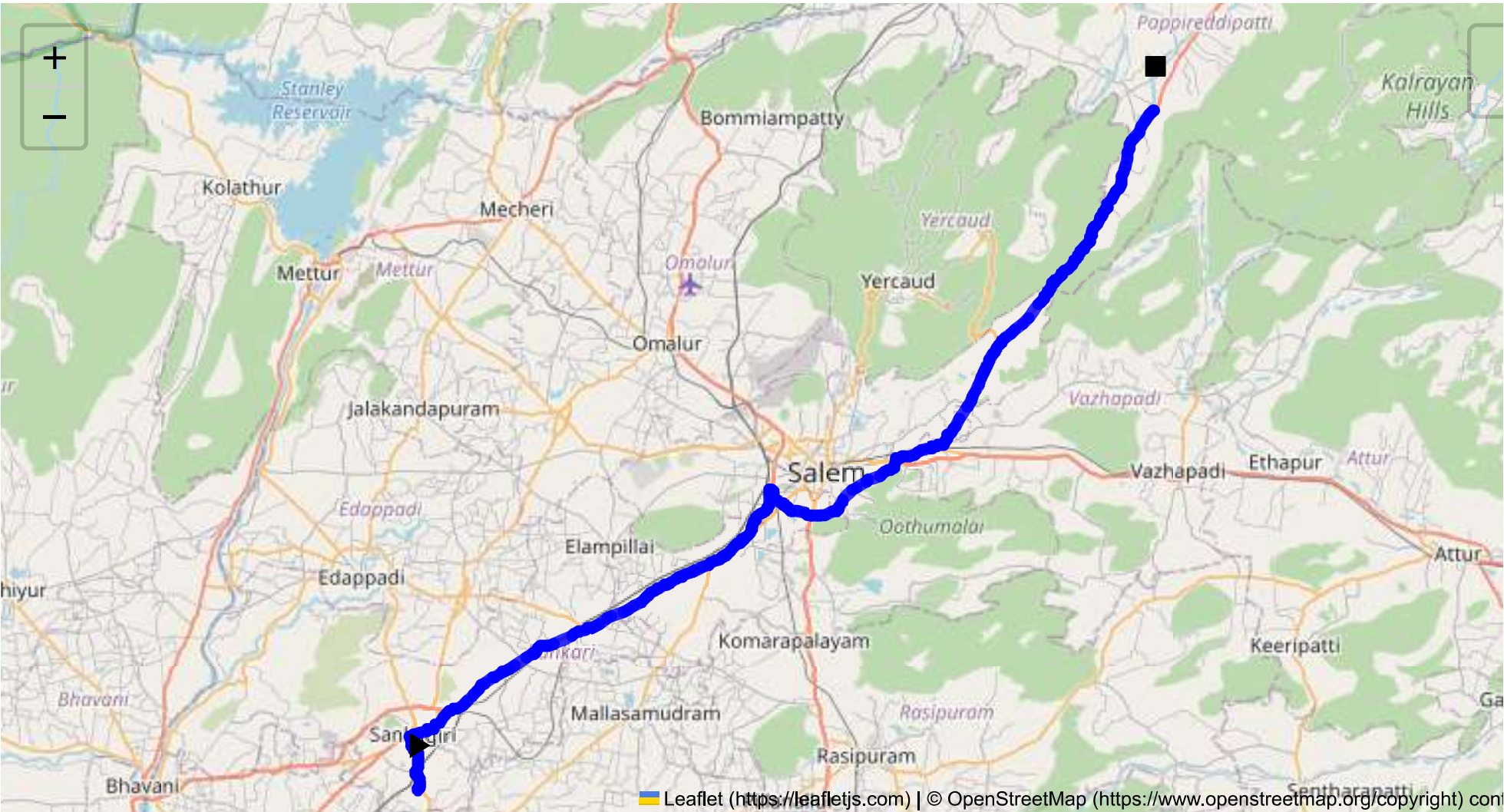
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 85.54 km
Estimated Duration: 2.2 hours
Adjusted Duration (Heavy Vehicle): 2.8 hours
Start: (11.4381, 77.8734)
End: (11.9004, 78.3847)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map

- The route from CVQF+23W (Sangagiri, Tamil Nadu) to W92M+5V (Alamelupuram, Tamil Nadu) is approximately 85.54 kilometers.
- The primary roads include state highways and local roads. The journey typically passes through both rural areas and smaller towns.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** The region generally experiences a tropical climate, with hot summers, mild winters, and a monsoon season typically from June to September.
- **Weather-Related Hazards:** Monsoon can cause heavy rainfall, leading to waterlogging and reduced visibility. During the summer, high temperatures may affect vehicle performance.

3. Traffic Patterns

- **Peak Hours:** Traffic is usually heavier in the early morning (7-9 AM) and evening (5-8 PM) around town areas due to local commuters.
- **Congestion-Prone Areas:** Town centers and intersections are more likely to experience congestion, especially near marketplaces and schools.

4. Road Quality and Infrastructure

- **Road Conditions:** Generally, state highways are well-maintained, but some local roads may have potholes or be narrow.
- **Infrastructure:** Signage may be limited in rural areas. Bridge and railway crossings are generally safe but require caution during rains.

5. Alternative Routes for Emergencies

- Alternative routes include taking different state highways that run parallel but may add extra distance or time. Using bypass roads for small towns can avoid congestion but requires more local knowledge.

6. Local Regulations Affecting Hazardous Material Transport

- Transport of hazardous materials is regulated under the Motor Vehicles Act, requiring specific permits and adherence to designated timings and routes.
- Nights are usually preferred for transporting hazardous materials to minimize risk.

7. Historical Incidents

- There have been sporadic incidents involving heavy vehicles, mainly minor accidents due to human error or mechanical failure. Major hazardous materials incidents are rare.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Include schools and water bodies. Ensure minimal impact by planning travel during student off-hours near schools.
- **Environmental Concerns:** Monitor closely for spills, especially near agricultural zones.

9. Communication Coverage

- **Dead Zones:** Some rural stretches may experience low mobile signals; it is recommended to equip vehicles with alternative communication devices.

10. Estimated Emergency Response Times

- **Rural Areas:** Expect longer response times (30-60 minutes) due to distance and road accessibility issues.
- **Town Areas:** Slightly quicker response times due to closer proximity to emergency services (15-30 minutes).

12. Summary of Risk Assessment

- **Overall Risk Level:** Moderate. While the main roads are typically safe and well-traveled, risks increase during monsoon seasons due to rain and potential flooding.
- **Key Recommendations:** Travel with awareness during adverse weather, stay updated on weather forecasts, ensure vehicle health checks before departure, carry alternative communication means, and maintain awareness of local traffic laws and regulations.

This report is intended to enhance the safety and preparedness of truck drivers transporting hazardous materials along this route. It is imperative to follow all safety guidelines and remain vigilant throughout the journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45000, 77.87213	30 KM/Hr
6	Turn	High	11.45028, 77.87202	15 KM/Hr
7	Turn	Medium	11.45309, 77.87325	30 KM/Hr
8	Turn	Medium	11.46400, 77.87323	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	Medium	11.46765, 77.87037	30 KM/Hr
10	Turn	High	11.47403, 77.86841	15 KM/Hr
11	Turn	High	11.49149, 77.89601	15 KM/Hr
12	Turn	High	11.49224, 77.89601	15 KM/Hr
13	Turn	Medium	11.61736, 78.10588	30 KM/Hr
14	Turn	Medium	11.61753, 78.10608	30 KM/Hr
15	Turn	Medium	11.64172, 78.11817	30 KM/Hr
16	Turn	Medium	11.64216, 78.11875	30 KM/Hr
17	Turn	Medium	11.64195, 78.12129	30 KM/Hr
18	Turn	Medium	11.64162, 78.12160	30 KM/Hr
19	Turn	Medium	11.64120, 78.12168	30 KM/Hr
20	Turn	Medium	11.63948, 78.12081	30 KM/Hr
21	Turn	Medium	11.66366, 78.20649	30 KM/Hr
22	Turn	Medium	11.67129, 78.22978	30 KM/Hr
23	Turn	High	11.67120, 78.23136	15 KM/Hr
24	Turn	Medium	11.67198, 78.23148	30 KM/Hr
25	Turn	Medium	11.67440, 78.23914	30 KM/Hr
26	Turn	Medium	11.80440, 78.33325	30 KM/Hr
27	Turn	Medium	11.83482, 78.35294	30 KM/Hr
28	Turn	High	11.90064, 78.38445	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
13	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
14	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
17	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
21	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
23	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
24	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
25	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
26	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
27	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
28	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
29	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
30	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
31	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
32	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
34	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
36	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
15	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
16	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
18	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
19	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
20	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
22	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
33	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
35	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

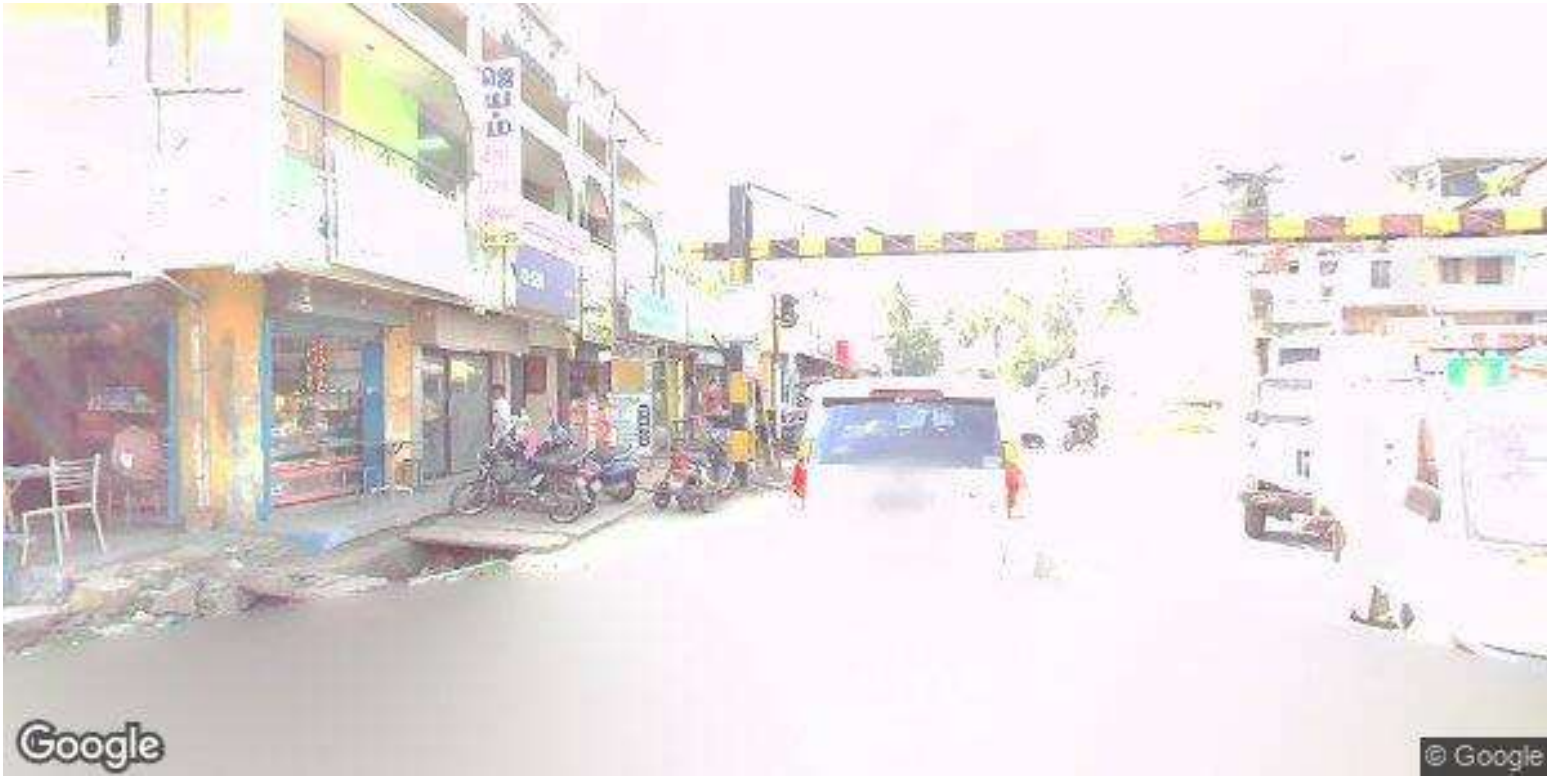
Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45000, 77.87213



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45309, 77.87325



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61753, 78.10608



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64172, 78.11817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64216, 78.11875



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64195, 78.12129



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64162, 78.12160



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64120, 78.12168



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63948, 78.12081



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66366, 78.20649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67129, 78.22978



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67120, 78.23136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67198, 78.23148



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67440, 78.23914



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80440, 78.33325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83482, 78.35294



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.90064, 78.38445

